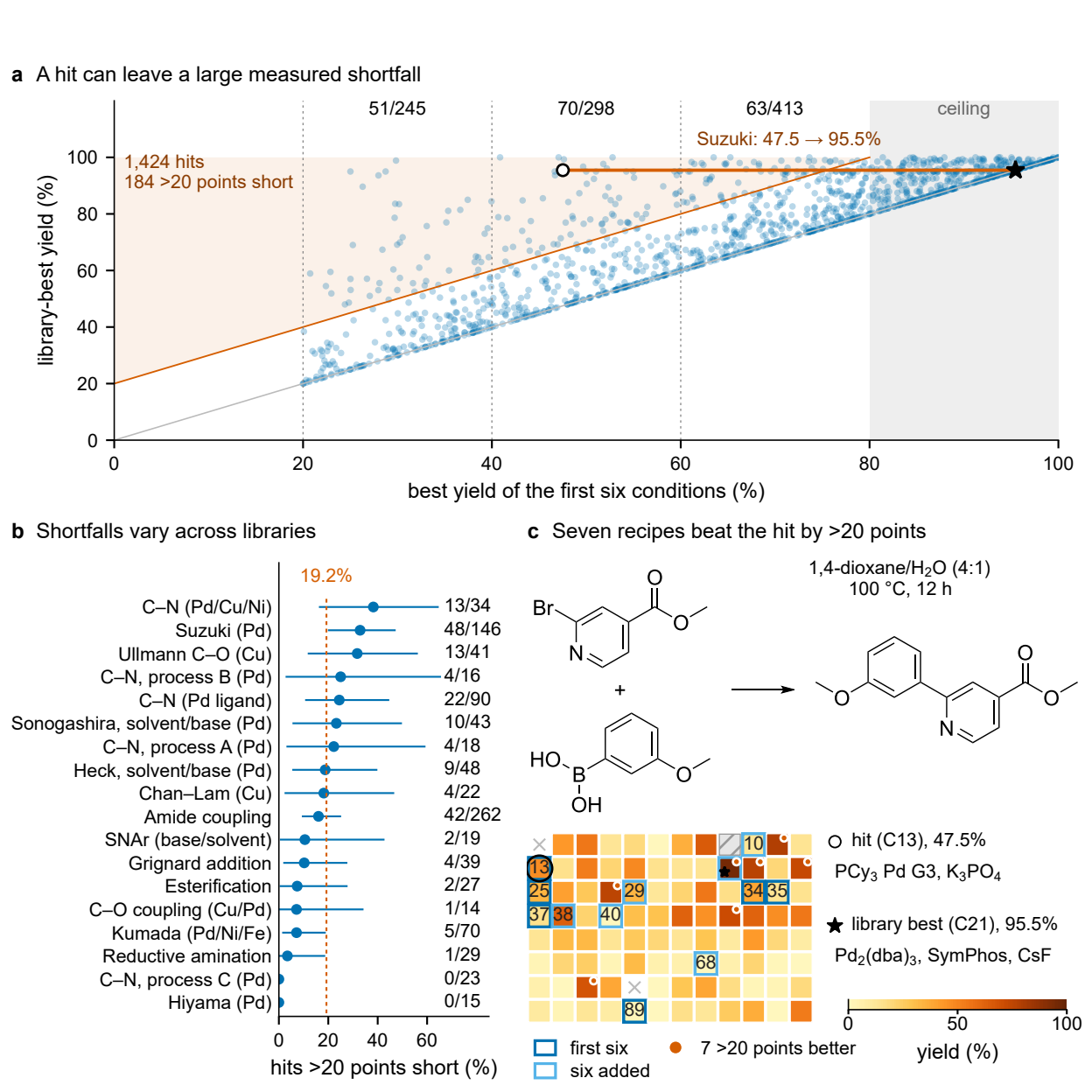

a A hit can leave a large measured shortfall
51/245
70/298
63/413
ceiling
Suzuki: 47.5 → 95.5%
1,424 hits
184 >20 points short
library-best yield (%)
best yield of the first six conditions (%)
b Shortfalls vary across libraries
19.2%
C–N (Pd/Cu/Ni) 13/34
Suzuki (Pd) 48/146
Ullmann C–O (Cu) 13/41
C–N, process B (Pd) 4/16
C–N (Pd ligand) 22/90
Sonogashira, solvent/base (Pd) 10/43
C–N, process A (Pd) 4/18
Heck, solvent/base (Pd) 9/48
Chan–Lam (Cu) 4/22
Amide coupling 42/262
SNAr (base/solvent) 2/19
Grignard addition 4/39
Esterification 2/27
C–O coupling (Cu/Pd) 1/14
Kumada (Pd/Ni/Fe) 5/70
Reductive amination 1/29
C–N, process C (Pd) 0/23
Hiyama (Pd) 0/15
hits >20 points short (%)
c Seven recipes beat the hit by >20 points
1,4-dioxane/H2O (4:1)
100 °C, 12 h
hit (C13), 47.5%
PCy3 Pd G3, K3PO4
library best (C21), 95.5%
Pd2(dba)3, SymPhos, CsF
first six
six added
7 >20 points better
yield (%)